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Lineární algebra – domácí úkol č. 2

Najděte všechna řešení následující soustavy rovnic:

$$\begin{aligned} 3x_1 + 5x_2 + 2x_3 + 3x_4 &= -3 \\ 2x_1 + 3x_2 + 0x_3 + x_4 &= -2 \\ x_1 + x_2 + 2x_3 + x_4 &= 1 \\ 4x_1 + 4x_2 + 3x_3 + x_4 &= 2 \end{aligned}$$

Převedu soustavu na rozšířenou matici soustavy:

$$\left(\begin{array}{cccc|c} 3 & 5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 1 & -2 \\ 1 & 1 & 2 & 1 & 1 \\ 4 & 4 & 3 & 1 & 2 \end{array}\right)$$

A dále pomocí Gaussovy-Jordanovy eliminace převedu tuto matici do redukovaného odstupňovaného tvaru:

$$\left(\begin{array}{cccc|c} 3 & 5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 1 & -2 \\ 1 & 1 & 2 & 1 & 1 \\ 4 & 4 & 3 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 2 & 3 & 0 & 1 & -2 \\ 3 & 5 & 2 & 3 & -3 \\ 4 & 4 & 3 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 2 & -4 & 0 & -6 \\ 0 & 0 & -5 & -3 & -2 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 4 & 2 & 2 \\ 0 & 0 & -5 & -3 & -2 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & -5 & -3 & -2 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 10 & 5 & 5 \\ 0 & 0 & -10 & -6 & -4 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 10 & 5 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -4 & -1 & -4 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right)$$

A tedy:

$$\begin{aligned} x_1 &= 1 \\ x_2 &= -1 \\ x_3 &= 1 \\ x_4 &= -1 \end{aligned}$$